DHIRENDRA JHA

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Professional Summary:

Software Engineer with 8+ years of experience designing backend systems, automation platforms, and scalable web applications using Python and Django. Experienced in distributed task processing, performance optimization, REST APIs, and offline-first systems, with a track record of delivering solutions ranging from enterprise automation to educational platforms serving 10,000+ users.

Skills:-

Backend: Python, Django, Django REST Framework, FastAPI, Golang
Frontend: JavaScript
Databases: PostgreSQL, MySQL, SQLite3
DevOps & Tools: Git, GitHub, GitLab, Docker (Basic), AWS EC2 (Basic)
Others: Raspberry Pi, Pandas, PySpark, Shell Scripting.

Key Achievements

• Built and deployed an offline LMS using Python, Django, and Raspberry Pi, serving 10,000+ rural students and improving academic performance by 50%.
• Optimized Django ORM queries, reducing data processing time from 1 hour to 5 seconds.

Personal Projects:

- LRU Cache (GoLang)

- Implemented a thread-safe **Least Recently Used (LRU) cache** using `sync.RWMutex`, doubly linked list and hash map, achieving **O(1)** insert, lookup, goroutines and eviction operations.

- TTLYD - Concurrent In-Memory TTL Cache Server (GoLang)

- **Designed and implemented a concurrent in-memory TTL cache in Go** using a hash map, doubly linked list, and min-heap to achieve efficient key lookups and expiration scheduling.
- **Implemented hybrid expiration strategies** combining lazy expiration with background eviction to balance lookup latency and memory reclamation.
- **Built a thread-safe cache** using `sync.RWMutex`, goroutines, channels, and `sync.WaitGroup` to support concurrent access and background processing.
- **Implemented graceful shutdown** using `context.Context`, ensuring background workers terminate cleanly while preserving in-flight operations.
- **Exposed the cache as an HTTP service (TTLYD)**, transforming the cache from a CLI utility into a reusable network-accessible service.

Work Experience (Official Projects)
EY GDS, Bangalore — Senior Consultant 2(September 2025 - Present)
Project: Distributed Sonarqube Issue Collector

- Designed and developed a **Python/Tkinter desktop application** to automate SonarQube issue collection and generate real-time comparison reports across multiple portfolios, streamlining release quality analysis.
- Currently designing and implementing a **Django based web version using Celery, Redis, and SQLite** to support asynchronous report generation and historical report management.
- Built the initial backend architecture for report generation and data persistence, with ongoing work on distributed task processing and interactive dashboards.

Accenture Solutions Pvt Ltd, Pune — Application Developer Team Lead (Aug 2024 – Aug 2025)
Project: Risk Tech Analysis

- **Designed and implemented Python automation scripts** to compare large datasets, calculate percentage differences, and validate risk model outputs, improving data consistency.
- **Automated dataset and lens provisioning using Azure CLI**, reducing environment setup time from several hours to a few minutes and minimizing manual effort.

Tech Stack: Python, Pandas, PySpark, GitLab.

Tata Consultancy Services, Mumbai — I.T. Analyst (Aug 2022 – July 2024)
Project: AES Cash processing.

- Developed a Django web application using **Python, Pandas, and OpenPyXL** to automate Excel data processing workflows, eliminating repetitive manual tasks and improving processing accuracy.
- **Implemented automated Excel data comparison and validation features using Pandas**, reducing manual effort by **80%** and accelerating report generation.

Tech Stack: Python, Django, Pandas, PostgreSQL, git, github.

Pratham Education Foundation, Mumbai — Software Engineer (July 2021 – July 2022)
Project: Offline Content Serving Mechanism Platform(Raspberry Pi)

- **Built and deployed a low-power Learning Management System (LMS) using Python, Django, and Raspberry Pi**, enabling offline content delivery in rural schools and supporting **10,000+ students**.
- **Improved student academic performance by 50%** through an offline-first digital learning platform optimized for low-resource environments.
- **Built a Python-based media conversion pipeline using FFmpeg** to automatically transcode diverse video and audio formats into standardized MP4 and MP3 outputs.

Tech Stack: Python, Django, Django REST Framework, Raspberry Pi, Shell Scripting, git, JavaScript, Jquery, Sqlite3.

Konsultera Solutions Pvt Ltd, Mumbai — Software Developer (Dec 2020 – June 2021)

Project: LAC Rebuild (Multi-tenant Assessment Platform)
- **Designed and implemented a scalable multi-tenant Django architecture**, allowing multiple educational institutions to manage students and learning content independently.
- **Optimized complex database queries using Django ORM subqueries**, improving data processing performance by over **700×** (1 hour to 5 seconds).

Tech Stack: Python, Django, PostgreSQL, PynamoDB, Javascript, Jquery

Pratham Education Foundation, Mumbai — Software Developer (Mar 2018 – Dec 2020)
- Developed multiple solutions for offline education using Raspberry Pi and LMS integration, one of which is Creating a Desktop Based application to work in correspondence to handle the data coming from Kolibri platform(offline education platform).
- Created and Configured Raspberry Pi as a standalone server to deliver Kolibri content and provide offline internet access in rural areas of india.

Tech Stack: Django, Django rest framework, JavaScript, Sqlite3, Raspberry PI, Linux, Shell Scripting, git, github, Raspbian OS.

Internships:-
Intern - Python/Django Developer, Insomniacs (Dec 2017 – Feb 2018)

Education:
B.E. (Information Technology)
Shree L.R. Tiwari College of Engineering, Mumbai (2012 – 2016) | CGPA: 6.47